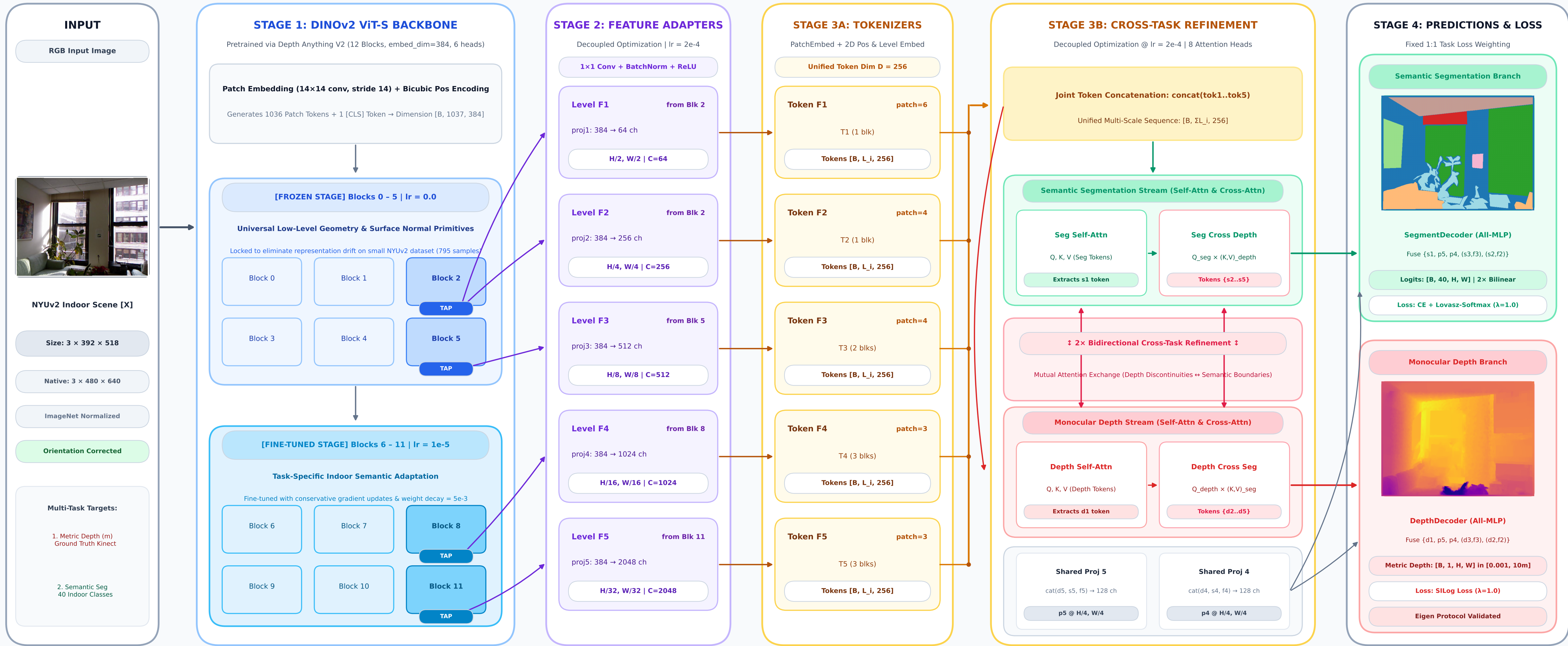


TwinForge: Decoupled Multi-Task Vision Architecture

Joint Monocular Depth Estimation & Semantic Segmentation with Early-Frozen DINOv2 ViT-S & Cross-Task Transformer Decoder

NYUv2 SOTA Benchmark: 0.5813 mIoU | 0.3650m Depth RMSE | 0.9112 61 | 24.91 GMac (122.7 GMac/Mpx) | 40.1 FPS (RTX 3060)



ARCHITECTURAL KEY & DESIGN PRINCIPLES:

[FROZEN FOUNDATION] Blocks 0-5 ($\text{lr}=0.0$): Locks universal low-level visual edge/gradient primitives; bypasses NYUv2 795-sample overfitting.

[FINE-TUNED BACKBONE] Blocks 6-11 ($\text{lr}=1e-5$): Cautiously adapts high-level indoor semantics & global layout without corrupting earlier representations.

[DECOUPLED OPTIMIZATION] Adapters & MultiHeadDecoder ($\text{lr}=2e-4$): $20 \times$ higher LR provides gradient capacity for multi-scale fusion.

[CROSS-TASK REFINEMENT] $2 \times$ Bidirectional Attention: Mutual exchange lets depth discontinuities sharpen segmentation boundaries.

Parameters: 45.74M (34.34M Train)

Compute: 24.91 GMac (122.7/Mpx)

Latency: 24.9ms / 40.1 FPS (RTX 3060)